

Page Navigation Techniques

Page Navigation=Taking user from 1 page to another page.

2 Ways

1. Client Pull

Taking the client to the next page in the NEXT request (coming all the way from client)

1.1 User takes some action --eg : clicking on a submit button or link & then client browser generates new URL to take user to the next page.

1.2 Redirect Scenario

User doesn't take any action. Client browser automatically generates new URL to take user to the next page.(next page can be from same web appln , or diff web appln on same server or any web page on any svr)

API of HttpServletResponse i/f

public void sendRedirect(String redirectLocation) throws IOException

becoz Redirecting a client involves network input/output operations means sending and receiving data over n/w

eg : For redirecting client from LoginServlet(/validate) to AdminServlet , in the NEXT request (/admin_page) , use

response.sendRedirect("admin_page");

What happens internally ?

WC -- discards resp buffer.

sends temp redirect response to the clnt

SC 302 | Headers : Location : admin_page | Body : EMPTY

Web browser --> generate a NEW redirect request

URL : http://host:port/ctx_path/admin_page ,method = GET

Meaning : you are navigating the clnt to the next page , in the NEXT request.

If the response already has been committed(pw flushed or closed) , this method throws(WC) an IllegalStateException.(since WC can't redirect the client after response is already committed)

2. Server Pull.

Taking the client to the next page in the SAME request.

Also known as resource chaining or request dispatching technique.

Client sends the request to the servlet / JSP. Same request can be chained to the next page for further servicing of the request.

Steps

1. Create Request Dispatcher object for wrapping the next page(resource --can be static or dynamic)

API of ServletRequest

javax.servlet.RequestDispatcher getRequestDispatcher(String path)

2.Forward scenario

API of RequestDispatcher

public void forward(ServletRequest rq,ServletResponse rs) rd.forward()

This method allows one servlet to do initial processing of a request and another resource to generate the response. (i.e division of responsibility)

Same as sendRedirect()

Uncommitted content in the response buffer is automatically cleared before the forward.

If the response already has been committed(pw flushed or closed) , this method throws an

IllegalStateException.

Limitation --only last page in the chain can generate dynamic response.

3. Include scenario

API of RequestDispatcher

```
public void include(ServletRequest rq,ServletResponse rs)
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Includes the content of a resource @run time (servlet, JSP page, HTML file) in the response. -- server-side includes.

Limitation -- The included servlet/JSP cannot change the response status code or set headers; any attempt to make a change is ignored.